

John Balci

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EDUCATION

Purdue University

B.S. in Electrical Engineering | GPA: 3.58/4.0

West Lafayette, IN

Aug. 2025 – May 2028 (Expected)

EXPERIENCE

Instrumentation and Hardware TRMC Intern

Summer 2026

Naval Test Wing Atlantic, Advanced Systems Group (Hired by ASRC Federal TRMCi)

Patuxent River, MD

- Programmed **TTC data acquisition hardware** via CDAUs and completed **pre-flight instrumentation checkout** on 2 E-2D Hawkeye aircraft for VX-20 Squadron.
- Diagnosed a pilot voice telemetry failure by **probing the telemetry panel with an oscilloscope**; isolated a GPS outage by verifying antenna power with a voltmeter, then **replaced the faulty antenna** to restore telemetry.
- Architected an IRIG-106 PCM frame synthesizer in Python using a **combinatorial solver** for PCM grids.
- Engineered a **2D Backtracking CSP solver** with forward checking and **6-metric heuristic** to place sensor data into PCM maps; integrated automated color-coded Excel exports, automating **20-40 hour process** to seconds.

Embedded Hardware & Avionics Undergraduate Researcher

Aug. 2025 – Present

Purdue Cubesat Lab, Electric Propulsion and Plasma Lab

West Lafayette, IN

- Synthesized 1V half-wave sinusoidal pulses (20 microsecond duration, 1Hz PRF) in **MATLAB** to characterize discharge cycles for a novel **pulsed-magnetoplasma dynamic** (PMPD) thruster in a 6U CubeSat.
- Engineered a precision capacitive gauging system** for propellant mass measurement, using CAPDAC to offset parasitic capacitance at a high-resolution 15pF dynamic range.
- Developed fringing field correction factors** for the plate model, for 0.23 mm displacement accuracy.

Founder, Principal Owner, Scrappr.site

Sept 2025 – Present

JEB Digital Ventures

Fairfax, VA

- Architected a **semantic search engine** using OpenAI text embeddings and weighted cosine similarity math to rank faculty profiles against research queries using OpenAlex API publication data **with 26 users**.
- Developed an **end-to-end AI outreach pipeline** turning CVs into personalized emails and cover letters.

Lead Propulsion Engineer

2021 – 2025

Caelus Rocketry

Alexandria, VA

- Led a 17-student propulsion & avionics team to build a liquid-fueled rocket engine through **cold flow tests**.

PROJECTS

PMPD Thruster Payload Interface Board | KiCad, STM32, Analog Design, I²C, Op-Amps

2026

- Designed a **4-channel data acquisition PCB** (65×95 mm, 2-layer, 2 oz Cu) for PMPD thruster ground testing, integrating pressure and flow sensors over a shared I²C bus to a NUCLEO-G431KB.
- Built an isolated **12 A Hall-effect current channel** (ACS725) and a single-rail **Rogowski coil op-amp integrator** biased at 1.65 V to capture fast discharge-pulse current.

Accumulator Capacitance Test Board | KiCad, FDC1004, I²C, Embedded C/C++

2026

- Designed a **35×25 mm 2-layer capacitance-to-digital PCB** (FDC1004) with a **driven shield** to measure propellant bladder displacement over a 2.0–1.5 mm plate gap, enabling low-flow mass-rate measurement for Isp.
- Derived a **parallel-plate model** converting capacitance to plate displacement and volume, calibrated against the empty-accumulator reference.

Gyroscopic Stabilized Pen | C++, OpenCV, ESP32-S3, Raspberry Pi 5, Git

2024 – 2025

- Built a closed-loop CV and sensor-fusion stabilization pen for users with essential tremor and cerebral palsy.

TECHNICAL SKILLS

EE: KiCad, LTSpice, MATLAB/Simulink, Circuit Analysis & Design, PCBs, MOSFETs, Verilog

Embedded: STM32, ESP32, Arduino, Raspberry Pi, Embedded C/C++, UART/SPI/I2C, Sensor Interfacing, Jetson

Instrumentation & Test: Telemetry Systems, Capacitive Measurement, Signal Synthesis, Data Acquisition

MCAD: Fusion 360, SolidWorks, Onshape, Autodesk Inventor, CNC Machining

Coding: Java, Python, C/C++, Swift, SQL (Postgres), JavaScript/Typescript, HTML/CSS, MATLAB